

TrES-3b

R-1048

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160530 · created 2026-07-30T09:27:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457538.8934 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.024
Transit depth (Rp/Rs)^2	1.32 +/- 0.54 [%]
Orbital Inclination (inc)	84.77 +/- 1.69
Airmass coefficient 1 (a1)	0.998 +/- 0.011
Airmass coefficient 2 (a2)	0.001 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	#2 - [555, 116]
Optimal Aperture	3.64
Optimal Annulus	5.16
Transit Duration (day)	0.0675 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.024 vs 0.16309 (deviation 1.39σ)
Duration fit vs published	0.0675 d vs 0.0567 d (deviation 0.77σ)
Mid-time O–C	2.2 min
Depth SNR	3.3
Residual scatter	1.14 %

Light curve

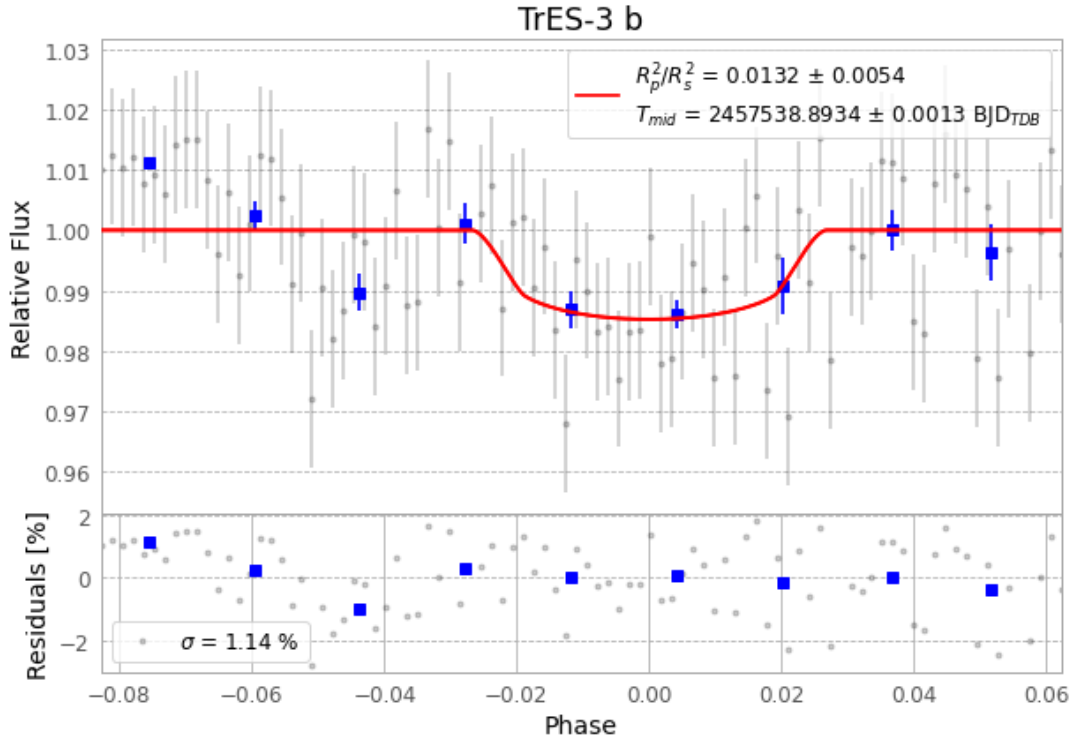


Figure 1 — FinalLightCurve_TrES-3 b_2016-05-29.png (SHA-256 69cc3b4b1568f4ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a32e8ae0730dd17...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), TRES-3160530064512.FITS → TRES-3160530111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160530.log (71ace7bf2eaa...), FinalLightCurve_TrES-3 b_2016-05-29.png (69cc3b4b1568...), FinalLightCurve_TrES-3 b_2016-05-29.csv (20bf701ee392...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 09:27Z.